

Hyojoon Park

PH.D. CANDIDATE | RESEARCH SCIENTIST (3D GRAPHICS, VISION, DEEP LEARNING)

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Introduction

- My research integrates **Machine Learning** with **Computer Graphics** and **Computer Vision**, building end-to-end systems that combine physics-based simulation, geometry, and image/video processing.
- My work spans **3D reconstruction** (face, body, hands), **XR (VR/AR)**, and **haptics**, with peer-reviewed publications in both Computer Graphics and Medical Imaging.
- Currently, I focus on **neural compression for 4D medical image and video**.

Education

University of Wisconsin-Madison

Wisconsin, USA

Ph.D. Candidate in Computer Sciences

Sep. 2021 - Expected Aug. 2026

- **Research area:** Synergistic integration of machine learning in physics-based simulation for computer graphics and medical imaging
- **Advisor:** Professor Eftychios Sifakis, **Co-advisor:** Professor Kevin Elceiri

Seoul National University

Seoul, Korea

M.S. in Mechanical Engineering

Mar. 2017 - Feb. 2019

- **Research area:** Rendering and transparent control of high-performance haptic system
- **Thesis:** Dental Simulator with Increased Z-width of Haptic Rendering (also presented at AsiaHaptics 2018)
- Selected for **Outstanding MS Thesis Presentation Award** [[M.S. thesis presentation](#)]
- **Advisor:** Professor Dongjun Lee

Technical University of Munich (TUM)

Munich, Germany

B.S. Exchange Student in Mechanical Engineering

Spring 2014

Korea University

Seoul, Korea

B.S. in Mechanical Engineering

Mar. 2010 - Feb. 2017

- Military service: Jun. 2011 - Mar. 2013

Work Experience

Google

Playa Vista, CA, USA

Student Researcher

Sep. 2025 - May 2026

- Developed a neural hair simulation research prototype for real-time XR graphics, focusing on efficient physics-based animation and publication-track technical results.

Nokia Bell Labs

New Providence, NJ, USA

Industrial Metaverse Intern

Jun. 2025 - Aug. 2025

- Built real-time 3D Gaussian Splatting scenes with precise physical registration (scale and orientation) and language features for natural language queries as part of "Empowering Digital Twins with Precise Physical Registration and Language Awareness." The project was selected as the 1st place winner among 100+ global interns.

NVIDIA

Santa Clara, CA, USA

Software Intern – Physics-Based Simulations

May 2024 - Aug. 2024

- Engineered an age-augmented 3D face reconstruction framework by generating synthetic training data spanning from infants to the elderly, covering a much wider age range. Developed a joint geometry-age inference model enabling real-time re-aging and de-aging of reconstructed faces.

University of Utah

Salt Lake City, UT, USA

Graduate Research Assistant

Sep. 2019 - Jun. 2021

- Architected the multi-camera calibration pipeline for "Capturing Detailed Deformations of Moving Human Bodies" (SIGGRAPH 2021), featuring robust checkerboard detection and bundle adjustment. The system captures 1,000+ uniquely labeled body points using a novel patterned suit and deep neural networks resilient to occlusion. [[code](#)] [[paper](#)]

Korea Institute of Science and Technology (KIST)

Seoul, Korea

Research Intern

Mar. 2019 - Aug. 2019

- Published "Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices" (IROS 2020), improving visual plausibility and precision of real-time 3D hand rendering in VR through motion retargeting and novel inverse kinematics constraints. [[paper](#)] [[video](#)]

Publications

- **Near-realtime Facial Animation by Deep 3D Simulation Super-Resolution**

Hyojoon Park, Sangeetha Grama Srinivasan, Matthew Cong, Doyub Kim, Byungsoo Kim, Jonathan Swartz, Ken Museth, Eftychios Sifakis

ACM Transactions on Graphics (TOG), 2024 (Presented at SIGGRAPH ASIA 2024) [\[paper\]](#) [\[code\]](#) [\[featured video\]](#)

- Achieves high-resolution facial animation 115x faster (at 18 FPS) than traditional methods while maintaining simulation quality, generalized to unseen expressions and dynamics.

- **Collagen Fiber Centerline Tracking in Fibrotic Tissue via Deep Neural Networks with Variational Autoencoder-based Synthetic Training Data Generation**

Hyojoon Park*, Bin Li*, Yuming Liu, Michael S. Nelson, Helen M. Wilson, Eftychios Sifakis, and Kevin W. Eliceiri (*equal contributions)

Medical Image Analysis, 2023 [\[paper\]](#)[\[code\]](#)

- Facilitates training of deep learning models with limited annotated data by generating diverse synthetic datasets via a novel property-controllable variational autoencoder.

- **Capturing Detailed Deformations of Moving Human Bodies**

He Chen, **Hyojoon Park**, Kutay Macit, and Ladislav Kavan

SIGGRAPH, 2021 [\[paper\]](#)[\[code\]](#)

- Enables accurate, detailed 3D reconstruction of moving human bodies using a novel motion capture system with a specially designed low-cost suit, utilizing deep neural networks and geometric algorithms.

- **Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices**

Hyojoon Park and Jung-Min Park

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020 [\[paper\]](#)[\[video\]](#)

- Enhances visual plausibility and precision of 3D hand rendering in VR using motion retargeting and novel hinge constraints for real-time inverse kinematics.

- **Dental Simulator with Increased Z-width of Haptic Rendering**

Hyojoon Park, Myungsin Kim, and Dongjun Lee

AsiaHaptics, 2018 [\[paper\]](#)[\[video\]](#)

- Achieves up to 10x greater maximum stiffness (Z-width) from virtual teeth using commercially available haptic devices.

- **Rigid-body Collaborative Manipulation among Remote Users with Wearable Cutaneous Haptic Interfaces**

Myungsin Kim, WonHa Lee, **Hyojoon Park**, Junghan Kwon, Yong-Lae Park, and Dongjun Lee

AsiaHaptics, 2018 [\[paper\]](#)[\[video\]](#)

- Introduces a remote multiuser collaboration system via wearable cutaneous haptic interfaces, leveraging passivity-based simulations for stable and realistic interactions.

- **Stretchable Skin-Like Cooling/Heating Device for Reconstruction of Artificial Thermal Sensation in Virtual Reality**

Jinwoo Lee, Heayoun Sul, Wonha Lee, Kyung Rok Pyun, Inho Ha, Dongkwan Kim, **Hyojoon Park**, Hyeonjin Eom, Yeosang Yoon, Jinwook Jung, Dongjun Lee, and Seung Hwan Ko

Advanced Functional Materials, 2020 [\[paper\]](#)

- Introduces a stretchable, bi-functional skin-like thermo-haptic (STH) device for VR, capable of precise cold and hot sensations with a single structure and 230% stretchability.

- **Wearable Cutaneous Haptic Interface with Soft Sensors and IMUs**

Yongjun Lee, Myungsin Kim, Yongseok Lee, **Hyojoon Park**, and Dongjun Lee

Korea Robotics Society Annual Conference, 2018

- **Design and Performance Evaluation of Wearable Haptic Interfaces**

WonHa Lee, Myungsin Kim, **Hyojoon Park**, and Dongjun Lee

International Conference on Control, Automation and Systems, 2018

Technical Skills

Languages	Python, C++, C, CUDA
Frameworks	PyTorch, JAX, OpenCV, NumPy/SciPy, OpenGL
Core Areas	3D Vision, Neural Rendering, Physics-based Simulation, Deep Learning, Computer Graphics, 3D Reconstruction, Data Compression
Tooling	Linux, Git, Docker, SLURM, CMake, $\text{\LaTeX}$

Teaching Assistant

Computer Graphics (CS559) *University of Wisconsin-Madison*

Spring 2022

Computer Graphics (CS559) *University of Wisconsin-Madison*

Fall 2021

Interactive Computer Graphics (CS6610) *University of Utah*

Spring 2021

Computer Graphics (CS4600) *University of Utah*

Fall 2020

System Analysis in Mechanical Engineering *Seoul National University*

Spring 2018